

Open Day 2026

This worksheet is accompanied by web apps that can be found here (<https://dundeemath.github.io/Admissions/>).

1 Introduction

The goal of this session is to explore how sounds can be represented mathematically.

- Why does the pitch of a string change as we adjust the tuning heads?
- Why does the pitch change when string length changes (e.g. by clamping strings on frets)?
- Why do some notes sound *nice* together?
- Why does an expensive musical instrument sound *nicer* than a cheaper one?

The worksheet is accompanied by a web app available at this [link](#).

You are not expected to have met some of the mathematical notation. This is covered on degree courses. But we can work with the solutions of the problem.

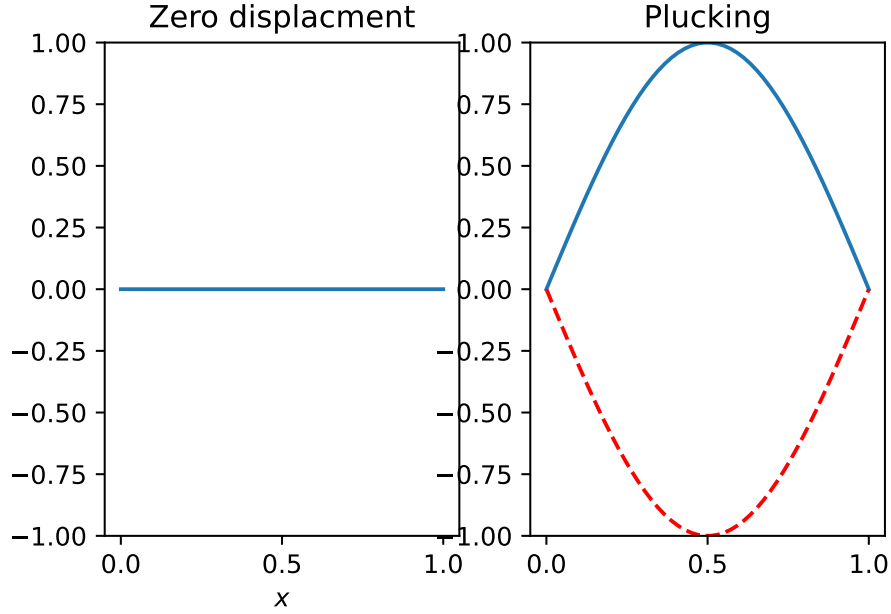
2 A mathematical model of a stretched string

2.1 Formulation

Consider a stretched string of length L (e.g. a guitar string). Suppose that the string holds a certain amount of tension, T , and has a certain line density, μ .

Thicker guitar strings will have a larger line density (μ is bigger). Similarly, we can increase tension in the string by twisting the tuning head.

Let $u(x, t)$ represent the displacement of the string at a given position, x , and time, t . When the string is at rest u is zero everywhere.



If the string is initially displaced (e.g. plucking a guitar string) we expect it to oscillate; this vibration generates the sound.

2.2 A governing equation

From a mathematical perspective, we can represent the string as a series of connected small elements. We can then describe the forces exerted on an element by its neighbours and use Newton's Second Law to connect forces to acceleration (i.e. movement).

A governing equation, known as the *wave equation*, can be written as

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad (1)$$

where the wave speed

$$c = \sqrt{\frac{T}{\mu}},$$

depends on the string tension and density.

As the boundaries of the string are fixed, the displacements there must be zero. This information is encoded via the *boundary conditions*

$$u(x=0, t) = 0, \quad u(x=L, t) = 0, \quad \forall \quad t.$$

We assume that the initial displacement is

$$u(x, t = 0) = f(x), \quad x \in [0, L].$$

Here $f(\cdot)$ is a function that represents the shape of the string at the instant it is plucked.

We also assume that the string is not moving initially, i.e. the initial velocity is zero

$$\left. \frac{\partial u}{\partial t} \right|_{(x, t=0)} = 0.$$

2.3 Solutions of the wave equation

The above system has solution known as a standing wave. It can be represented as

$$u(x, t) = \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi ct}{L}\right), \quad n = 1, 2, \dots \quad (2)$$

A formula for the frequency of a vibration (i.e. pitch) is given by

$$f = \frac{n}{2L} \sqrt{\frac{T}{\mu}}, \quad n = 1, 2, \dots \quad (3)$$

2.4 Preliminary exercises

1. Suppose we were studying a simpler differential equation

$$\frac{d^2 u}{dx^2} = 0 \quad (4)$$

To show that

$$u(x) = x$$

is a solution, differentiate once to get

$$\frac{du}{dx} = 1.$$

Differentiate again

$$\frac{d^2 u}{dx^2} = 0.$$

Hence $u(x) = x$ is a solution to the equation.

Can you show that $u(x) = x^2$ is a *not* a solution to Equation 4?

2. Sketch the function $\sin(x)$ for $x \in [0, 2\pi]$. On the same graph sketch $\sin(2x)$. Can you identify the local maxima and minima on the sketch?
3. Sketch the solution of Equation 2 at different values of t (e.g. try $t = 0$ and $t = 1$). You can assume that $L = 1, \mu = 1$ and $T = 1$.

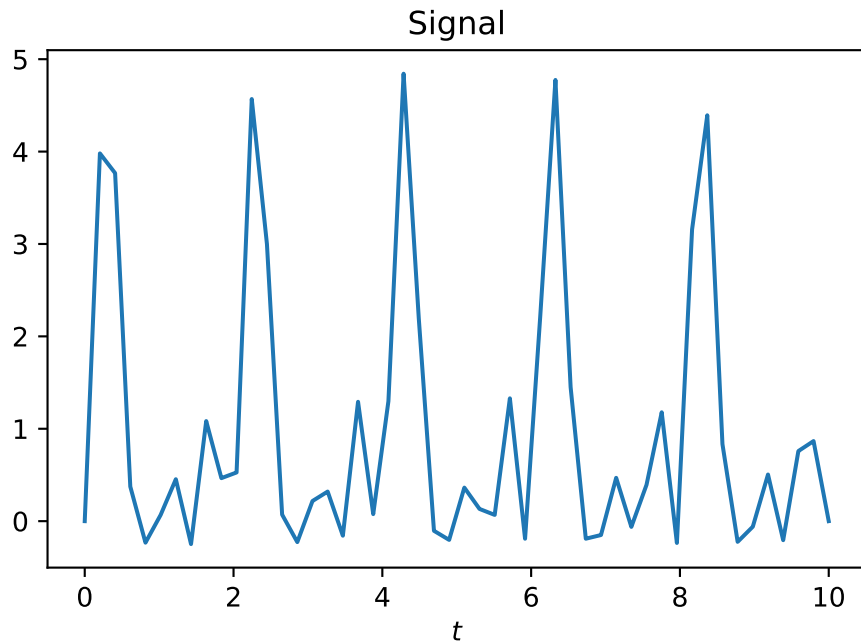
2.5 Exercises

Now let's return to the wave equation.

1. Verify that Equation 2 is a solution to Equation 1. Hint: substitute the solution into the governing equation.
2. Use Equation 3 to explore how the pitch of a note depends on model parameters. For example, sketch the formula for the frequency against μ (holding the others at fixed values, e.g. 1).
3. To double the frequency of a note using the tuning head (i.e. move up an octave), what factor do I have to increase the tension by (don't try this on your instrument in standard tuning!).
4. To explore other solutions of the wave equation follow this [link](#).

2.6 Signal analysis

Suppose we record an audio signal and it looks like the time series below.



There is a very nice branch of mathematics called Fourier series which shows that an (almost) arbitrary signal can be written as a sum of different sine waves. In the app we will explore how the frequencies in a signal can be represented in a *spectrogram*.